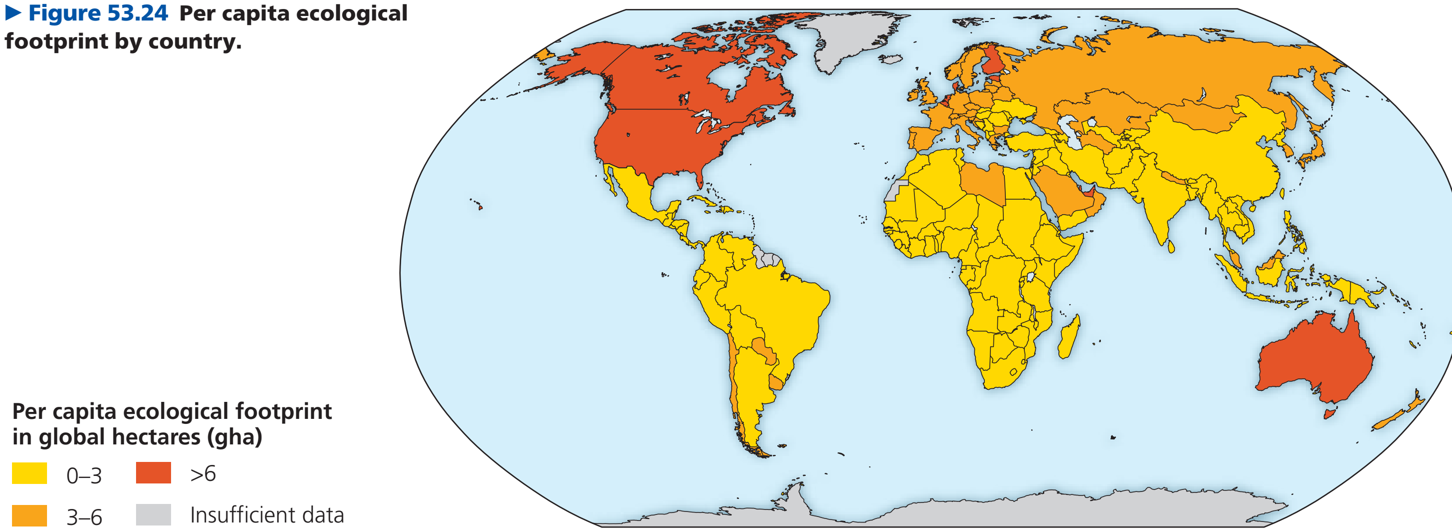
▶ **Figure 53.24 Per capita ecological footprint by country.**



Earth has a total of 11.9 billion gha of productive land. How many people could Earth support sustainably if the average ecological footprint were 8 gha per person (as in the United States)?

to produce all the resources it consumes and to absorb all the waste it generates.

What is a sustainable ecological footprint for the entire human population? One way to estimate this footprint is to add up all the ecologically productive land on the planet and divide by the size of the human population. Typically, this estimate is made using *global hectares*, where a global hectare (gha) represents a hectare of land or water with a productivity equal to the average of all biologically productive areas on Earth (1 hectare = 2.47 acres). This calculation yields an allotment of 1.7 gha per person—the benchmark for comparing actual ecological footprints. Anyone who consumes resources that require more than 1.7 gha to produce is using an unsustainable share of Earth's resources, as is the case for the citizens of many countries **(Figure 53.24)**. For example, a typical ecological footprint for a person in the United States is 8 gha. Globally, the average footprint is 2.7 gha per person, more than a 50% overshoot of the sustainable use (1.7 gha per person) of Earth's resources.

Our impact on the planet can also be assessed using currencies other than area, such as energy use. Average energy use differs greatly across different regions of the world **(Figure 53.25)**. A typical person in the United States, Canada, or Norway consumes roughly 30 times the energy that a person in central Africa does. Moreover, fossil fuels, such as oil, coal, and natural gas, are the source of 80% or more of the energy used in most developed nations. As Concept 56.4

▶ **Figure 53.25 The uneven electrification of the planet.** This composite image taken from space of Earth's surface at night illustrates the varied density of electric lights worldwide, one aspect of energy use by humans.